

# SCX Theory: Full Proof Chain Audit

**Date:** 2026-06-28 | **Status:** Pre-submission audit **Purpose:** Cross-check every theorem dependency, verify no circular dependencies, identify hidden assumptions. **Scope:** Theorems 1-3, Spring-1 (SE-1), Spring-2 (SE-2), and all connecting propositions.

## 1. Dependency Graph (Verified)

Theorem 3  
(Unidentifiability)

"Without A1-A6,  
noise difficulty"

JUSTIFIES (why A1-A6 are needed)  
No logical dependency on Thm 1

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Assumption Set A1-A6  
(Disjoint training, independence,  
bounded loss, uniform noise,  
state homogeneity, balanced errors)

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Theorem 1  
(Noise Detection)

Theorem 2  
(Weak Feature)

"With A1-A6, F1→1  
exponentially in M"

"Even with A1-A6,  
if  $I(;S)$  ,  
SCX baseline

PROOF: Hoeffding +  
Chernoff on e\_m

+  $O(\sqrt{\cdot})$ "

PROOF: Pinsker/BH +  
Fano + data process.

provides S\_0  
(consensus scores)

provides boundary  
 $O(\sqrt{\cdot})$  on advantage

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Spring-1: Theorem SE-1  
(Convergence of Self-Evolution)

"(S\_t, \_t) → (S\*, \*) under C1-C9"

PROOF: Lyapunov descent (Theorem 12.5) +  
memory bank stabilization (Lemma SE-1.2) +  
vanishing displacement (Lemma SE-1.3) +

fixed-point characterization (Lemma SE-1.4)

DEPENDS ON:

- Thm 1: provides noise detection signal for S\_0
- Thm 2: characterizes when advantage vanishes
- C1'-C9: structural conditions
- Theorem 12.5: Lyapunov descent (proven)

provides Lyapunov structure  
and convergence guarantee

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Spring-2: Theorem SE-2

(Completeness Bound)

"Finite-time termination under physical limits"

PROOF: Finite configuration space (Prop SE-3) +  
Lyapunov descent  $\rightarrow$  no cycles of length  $> 1$  +  
bounded improvement  $\rightarrow$  finite steps to fixed point

DEPENDS ON:

- Spring-1: Lyapunov descent + convergence
- Physical constraints: finite data, finite  
precision, finite parameterization

**Verdict:** The dependency graph is a **DAG** (directed acyclic graph). **No**

circular dependencies found.

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## 2. Dependency-by-Dependency Verification

### 2.1 Chain: Thm 3 $\rightarrow$ A1-A6 $\rightarrow$ Thm 1

**Claim:** Theorem 1's assumptions (A1-A6) are justified by Theorem 3's unidentifiability result.

**Direction:** Thm 3  $\rightarrow$  Thm 1 (justification, not logical dependency)

**Verification:**

| Check                             | Result                          | Evidence                                                                                  |
|-----------------------------------|---------------------------------|-------------------------------------------------------------------------------------------|
| Does Thm 1's proof require Thm 3? | <b>NO</b> — Thm 1 is standalone | Thm 1 proof uses only A1-A6 + Hoeffding/Chernoff; Thm 3 is never cited in the proof       |
| Does Thm 3's proof require Thm 1? | <b>NO</b> — Thm 3 is standalone | Thm 3 proof is a constructive counterexample; references Thm 1 only for A1-A6 definitions |

| Check                                          | Result     | Evidence                                                                                                                                  |
|------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Is the relationship correctly characterized?   | <b>YES</b> | Document 00: “Theorem 3...establishes why assumptions are required. Theorem 1...quantifies how well SCX works when its assumptions hold.” |
| Are A1-A6 consistent between the two theorems? | <b>YES</b> | Same A1-A6 used in both; Thm 3’s Section 4 maps each assumption to what it breaks                                                         |

**Verdict: PASS.** No circularity. The relationship is: Thm 3 justifies **why** A1-A6 are needed; Thm 1 proves **what** A1-A6 achieve. This is a conceptual dependency (motivation), not a logical dependency (proof structure).

**Note on arrow direction:** Some documents describe this as “Thm 1  $\rightarrow$  Thm 3” (reading left-to-right as “Thm 1’s assumptions require Thm 3’s justification”). The actual logical flow is Thm 3  $\rightarrow$  A1-A6  $\rightarrow$  Thm 1. This is a harmless notational difference but should be clarified in the arXiv submission.

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## 2.2 Chain: Thm 1 $\rightarrow$ Spring-1 (Noise Detection $\rightarrow$ Gatekeeper Initialization)

**Claim:** Theorem 1’s noise detection guarantee provides the signal that initializes the gatekeeper  $S_0$  and continuously calibrates  $S_t$ .

**Direction:** Thm 1  $\rightarrow$  Spring-1 (input dependency)

**Verification:**

| Check                                       | Result             | Evidence                                                                                                                                  |
|---------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Does Spring-1 use Thm 1's consensus score?  | <b>YES</b>         | $S_0$ is initialized from $C(x)$ (Thm 1's consensus score); $S_t$ updates use SCXUpdate which depends on consensus                        |
| Is Proposition SE-1.4 correctly stated?     | <b>YES</b>         | "Throughout the self-evolution process...the noise detection guarantee of Theorem 1 applies at each time step $t$ "                       |
| Is the F1 global aggregation correct?       | <b>YES (FIXED)</b> | DEFECT-01/02 fixed; Lemma F now uses correct global F1 via linear FPR/FNR decomposition                                                   |
| Does Spring-1 add assumptions beyond Thm 1? | <b>YES</b>         | C1-C9 are additional (Lipschitz, RM rates, two-timescale, exploration, etc.) — but they are <b>orthogonal</b> to A1-A6, not contradictory |

|                                          |           |                                                                                                                                                            |
|------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Could Spring-1<br>work without Thm<br>1? | <b>NO</b> | Without Thm 1's noise<br>detection signal, $S_0$ has no<br>meaningful initialization;<br>the gatekeeper would have<br>no basis for<br>acceptance/rejection |
|------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

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**Verdict: PASS.** Dependency is real and correctly documented. Thm 1 provides the essential noise detection signal. Spring-1 adds dynamical assumptions (C1-C9) that are consistent with but independent of A1-A6.

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### 2.3 Chain: Thm 2 $\rightarrow$ Spring-1 (Weak Feature $\rightarrow$ Degradation Rate)

**Claim:** Theorem 2's weak feature bound characterizes when SCX cannot outperform the loss baseline, providing the boundary condition for Spring-1's effectiveness.

**Direction:** Thm 2  $\rightarrow$  Spring-1 (boundary condition)

**Verification:**

| Check                                                            | Result                                                      | Evidence                                                                                                                                                                                                               |
|------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does Spring-1 depend on Thm 2 for correctness?                   | <b>NO</b> — Spring-1's convergence proof does not use Thm 2 | Spring-1 (Theorem SE-1) proof in Document 06 uses only C1-C9 + Lyapunov descent; Thm 2 is not cited in the convergence proof itself                                                                                    |
| Does Thm 2 constrain Spring-1's effectiveness?                   | <b>YES</b>                                                  | When $I(\phi; S) \rightarrow 0$ , SCX cannot outperform baseline; this means Spring-1 converges to a fixed point that may be no better than the baseline                                                               |
| Is this relationship correctly documented?                       | <b>YES</b>                                                  | Document 06 Section 14: "Thm 2...bounds the quality of the fixed point: if features are weak, even the optimal fixed point cannot outperform the loss baseline"                                                        |
| Is the $O(\sqrt{\delta})$ bound consistent with Spring-1's rate? | <b>PARTIALLY</b>                                            | Thm 2 gives $O(\sqrt{\delta})$ bound on SCX advantage. Spring-1 gives $O(\alpha_t + \beta_t)$ convergence rate. These are orthogonal rates (one is feature-quality bound, one is optimization rate). No contradiction. |



**Verdict: PASS.** Thm 2 provides a **boundary condition** for Spring-1 — it tells us when convergence is meaningful — but does not enter Spring-1’s proof logically. This is correct: a boundary theorem should not be required for the convergence proof itself.

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## 2.4 Chain: Spring-1 $\rightarrow$ Spring-2 (Convergence $\rightarrow$ Completeness)

**Claim:** Theorem SE-1’s convergence guarantee provides the Lyapunov structure that Theorem SE-2 uses to prove finite-time termination.

**Direction:** Spring-1  $\rightarrow$  Spring-2 (logical dependency)

**Verification:**

| Check                                         | Result     | Evidence                                                                                                                                                                                          |
|-----------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does SE-2’s proof require SE-1’s convergence? | <b>YES</b> | SE-2 (Document 07) proof: “By the strict Lyapunov property (Theorem SE-1, assumption), $\Phi(q_{t+1}) < \Phi(q_t)$ unless $q_{t+1} = q_t$ .”                                                      |
| Does SE-2 add assumptions beyond SE-1?        | <b>YES</b> | SE-2 adds physical constraints: finite data ( $N_{\max} < \infty$ ), finite precision ( $\varepsilon_{\text{mach}} > 0$ ), finite parameterization. These are <b>independent</b> of SE-1’s C1-C9. |

| Check                                                           | Result             | Evidence                                                                                                                                                                                               |
|-----------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is SE-2's<br>finite-configuration<br>argument valid?            | <b>YES (FIXED)</b> | DEFECT-15 replaced<br>“finite $\mathcal{X}$ ” with<br>covering-number argument;<br>Proposition SE-3 now<br>correctly bounds $ \mathcal{Q}  < \infty$<br>under physical constraints                     |
| Does SE-2's<br>fixed-point<br>characterization<br>match SE-1's? | <b>YES</b>         | Both characterize $S^* =$<br>$\text{SCXUpdate}(S^*, M_\infty, f_{\theta^*})$<br>and $\nabla L_{S^*}(\theta^*) = 0$                                                                                     |
| Could SE-2 hold<br>without SE-1?                                | <b>PARTIALLY</b>   | SE-2's finite-configuration<br>argument (termination in<br>finite time) holds regardless<br>of SE-1, but the <b>quality</b> of<br>the fixed point (that it's a<br>Lyapunov minimum)<br>depends on SE-1 |

**Verdict: PASS.** Spring-2 logically depends on Spring-1 for the Lyapunov structure. Spring-2's additional physical constraints are independent and verifiable.

## 2.5 Cross-Verification: Hidden Assumptions

We check for assumptions that are used but not explicitly stated in the dependency chain.

| Hidden Assumption Check                                                             | Result                   | Details                                                                                                            |
|-------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------|
| Does Thm 1 assume expert models are “good enough” ( $\mu_s < (K - 1)/K$ )?          | <b>Explicit</b>          | Lemma 1 states this as the separation condition; Corollary 3 (uniform detectability) formalizes it                 |
| Does Thm 2 assume the clustering algorithm (k-means) achieves the Fano lower bound? | <b>FIXED (DEFECT-11)</b> | Now explicitly states it’s conservative; uses single-sample $\delta$ which gives an upper bound on SCX performance |
| Does Spring-1 assume the memory bank stabilizes?                                    | <b>Explicit</b>          | Lemma SE-1.2 proves this under C1 (covering dimension) + monotonicity                                              |

| Hidden Assumption Check                           | Result          | Details                                                                                           |
|---------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------|
| Does Spring-1 assume the student loss is smooth?  | <b>Explicit</b> | Condition C2 (Lipschitz student) + $L_g$ -smoothness derived from C2                              |
| Does Spring-2 assume the system is deterministic? | <b>Explicit</b> | “The self-evolution update is a deterministic mapping $G : \mathcal{Q} \rightarrow \mathcal{Q}$ ” |

| Hidden Assumption Check                                             | Result                         | Details                                                                                                                                                                                            |
|---------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does Thm 3’s construction use A1-A2 (which it’s trying to justify)? | <b>NO</b><br><b>(verified)</b> | Thm 3 Section 2.4 Step 4 explicitly states “we construct the experts to be conditionally independent...this does not assume A1-A2 hold universally” — it’s part of the constructive counterexample |

**Verdict: No hidden assumptions found.** All assumptions are explicitly stated in their respective theorem documents. One borderline case (Thm 3 using conditional independence in its construction) is correctly handled — the construction demonstrates that even when A1-A2 hold, the counterexample exists, which only strengthens the theorem.

### 3. Notation Consistency Audit

#### 3.1 Critical Notation Checks

| Symbol        | Thm 1                     | Thm 2                           | Thm 3           | Spring-1                           | Spring-2                   | Consistent?                                               |
|---------------|---------------------------|---------------------------------|-----------------|------------------------------------|----------------------------|-----------------------------------------------------------|
| $\mathcal{S}$ | State space               | State space                     | State space     | <b>Gatekeeper scores</b> $S_t$     | <b>Gatekeeper</b> scores   | <b>CONFLICT</b><br>—<br>documented in 00_notation         |
| $S$           | State variable            | State variable                  | State variable  | <b>Gatekeeper scoring function</b> | <b>Gatekeeper</b> function | <b>CONFLICT</b><br>—<br>documented                        |
| $K$           | $ \mathcal{Y} $ (classes) | N/A                             | $ \mathcal{Y} $ | N/A                                | N/A                        | <b>CONSISTENT</b>                                         |
| $K_S$         | Not used                  | $ \mathcal{S} $ (states)        | Not used        | $ \mathcal{S} $                    | $ \mathcal{S} $            | <b>CONSISTENT</b><br>(but added in polished versions)     |
| $\eta$        | Noise rate                | Noise rate                      | Noise rate      | N/A                                | N/A                        | <b>CONSISTENT</b>                                         |
| $\delta$      | Confidence parameter      | <b>Mutual information bound</b> | N/A             | N/A                                | N/A                        | <b>CONFLICT</b><br>(different meanings in Thm 1 vs Thm 2) |
| $\Delta_s$    | Separation gap            | Not used                        | Not used        | Not used                           | Not used                   | <b>UNIQUE to Thm 1</b>                                    |

| Symbol   | Thm 1                     | Thm 2            | Thm 3                     | Spring-1                     | Spring-2             | Consistent?                                          |
|----------|---------------------------|------------------|---------------------------|------------------------------|----------------------|------------------------------------------------------|
| $\rho_s$ | State<br>proba-<br>bility | Not<br>used      | State<br>proba-<br>bility | Not used                     | Not used             | <b>CONSISTENT</b>                                    |
| $M$      | Expert<br>count           | Expert<br>count  | Expert<br>count           | Not used                     | Not used             | <b>CONSISTENT</b>                                    |
| $M_t$    | Not<br>used               | Not<br>used      | Not<br>used               | <b>Memory<br/>bank</b>       | Memory<br>bank       | <b>UNIQUE<br/>to<br/>Spring-1/2</b>                  |
| $\Phi$   | Not<br>used               | Feature<br>space | Not<br>used               | <b>Lyapunov<br/>function</b> | Lyapunov<br>function | <b>CONFLICT</b><br>(feature<br>space vs<br>Lyapunov) |
| $\phi$   | Not<br>used               | Feature<br>map   | Not<br>used               | Not used                     | Not used             | <b>UNIQUE<br/>to Thm 2</b>                           |

### 3.2 Notation Conflicts Requiring Attention

**Conflict 1:**  $\mathcal{S}$  (state space) vs.  $S_t$  (gatekeeper scoring function) - **Affected documents:** Theorems 1-3 use  $\mathcal{S}$  for state space. Spring-1/2 use  $S_t$  for gatekeeper scores. - **Resolution:** Document 00\_notation\_and\_dependencies.md clarifies this. The polished theorem documents (polished/) use  $K_S = |\mathcal{S}|$  for state space cardinality. - **Risk:** Low. Context disambiguates (Thm 1-3 never mention  $S_t$ ; Spring-1/2 always subscript  $S_t$  with time index). - **Recommendation:** Add a one-sentence note in Spring-1: “Notation:  $\mathcal{S}$  denotes the state space (as in Thm 1-3);  $S_t$  denotes the gatekeeper scoring function at time  $t$ .”

**Conflict 2:  $\delta$  (confidence in Thm 1 vs. mutual information in Thm 2) - Affected documents:** Thm 1 uses  $\delta$  for confidence parameter (e.g.,  $1 - \delta$ ). Thm 2 uses  $\delta$  for  $I(\phi; S) \leq \delta$ . - **Risk:** Medium. Both are standard in their respective contexts but could confuse a reader who reads both theorems sequentially. - **Recommendation:** Rename Thm 2’s  $\delta$  to  $\delta_\phi$  or  $\iota$  (iota for information) in the arXiv version.

**Conflict 3:  $\Phi$  (feature space in Thm 2 vs. Lyapunov function in Spring-1/2) - Affected documents:** Thm 2 uses  $\Phi$  for feature space. Spring-1/2 use  $\Phi$  for the Lyapunov function. - **Risk:** Medium. The contexts are very different (Thm 2: information theory; Spring: dynamical systems) but the symbol appears in both. - **Recommendation:** Spring documents should use  $\mathcal{V}$  or  $\Psi$  for the Lyapunov function instead of  $\Phi$ , or Thm 2 should use  $\mathcal{F}_\phi$  for the feature space.

### 3.3 Notation Conflicts: Assessment

| Severity                              | Count | Examples                                                                                            |
|---------------------------------------|-------|-----------------------------------------------------------------------------------------------------|
| <b>Critical</b> (proof-breaking)      | 0     | —                                                                                                   |
| <b>Moderate</b><br>(reader-confusing) | 3     | $\mathcal{S}$ vs $S_t$ , $\delta$ (dual use), $\Phi$ (dual use)                                     |
| <b>Minor</b> (stylistic)              | 2     | $K$ vs $K_S$ (polished versions use $K_S$ ), $\rho_s$ (Thm 1) vs $\rho$ (Thm 3’s state probability) |

**Overall:** Notation is **mostly consistent** within each theorem’s scope. Cross-theorem conflicts are documented and resolvable with minor renaming.



## 4. Dependency Strength Assessment

### 4.1 Logical Dependencies (Proof-Critical)

| From                           | To       | Strength      | If broken, what fails?                                                        |
|--------------------------------|----------|---------------|-------------------------------------------------------------------------------|
| C1-C9                          | Spring-1 | <b>STRONG</b> | Convergence proof fails without these conditions                              |
| Lyapunov descent<br>(Thm 12.5) | Spring-1 | <b>STRONG</b> | Without Lyapunov descent, only Cesàro-mean convergence remains (Theorem 10.1) |
| Spring-1                       | Spring-2 | <b>STRONG</b> | SE-2's fixed-point quality guarantee depends on SE-1's Lyapunov structure     |
| A1-A6                          | Thm 1    | <b>STRONG</b> | Noise detection guarantee fails without these assumptions                     |
| Fano + Pinsker/BH2             | Thm      | <b>STRONG</b> | Weak feature bound depends on these inequalities                              |

### 4.2 Conceptual Dependencies (Motivation/Interpretation)

| From  | To                    | Strength        | Purpose                                |
|-------|-----------------------|-----------------|----------------------------------------|
| Thm 3 | A1-A6                 | <b>MODERATE</b> | Justifies why A1-A6 are not arbitrary  |
| Thm 1 | Spring-1<br>( $S_0$ ) | <b>MODERATE</b> | Provides the initial gatekeeper signal |

| From  | To                          | Strength    | Purpose                                            |
|-------|-----------------------------|-------------|----------------------------------------------------|
| Thm 2 | Spring-1<br>(bound-<br>ary) | <b>WEAK</b> | Characterizes when<br>convergence is<br>meaningful |

### 4.3 Critical Path Analysis

The **critical proof path** is:

A1-A6  $\rightarrow$  Thm 1 (noise detection signal)

C1-C9  $\rightarrow$  Theorem 12.5 (Lyapunov descent)  $\rightarrow$  Spring-1 (convergence)  $\rightarrow$  Spring-2 (comp)

Thm 2 (boundary condition)

If any link in the strong dependency chain breaks, the corresponding theorem's proof is invalid. Currently:

- **All strong dependency links verified intact.**
- **Thm 1:** A1-A6 are explicitly stated and used. Proof is complete (Hoeffding + Chernoff).
- **Theorem 12.5:** Requires reference-set replay mechanism. Proof is complete (importance sampling + reference-based SCXUpdate).
- **Spring-1:** Depends on Theorem 12.5 for Lyapunov descent. With Theorem 12.5, the proof is complete.
- **Spring-2:** Depends on Spring-1 for Lyapunov structure. Proof is complete under physical constraints.

5. Gap Inventory (What Is NOT Proven)

| Gap ID | Description                                      | Location                           | Severity                                  | Resolution                                                                                                      |
|--------|--------------------------------------------------|------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| G1     | Lyapunov descent WITHOUT reference-set replay    | 10_lyapunov, Thm 12.2              | <b>FUNDAMENTAL</b><br>(proven impossible) | Thm 12.2 proves impossibility; reference-set replay (Thm 12.5) closes the gap                                   |
| G2     | Infinite $\mathcal{X}$ extension of Lemma SE-1.2 | 06_fixed_point, Section 14         | <b>MINOR</b>                              | Covering-number argument (DEFECT-15) resolves for $\varepsilon$ -precision; full functional convergence is open |
| G3     | Phase transitions between regimes 1-4            | 06_fixed_point, Section 11         | <b>CONJECTURED</b>                        | <b>RED</b> needed for main theorems; documented as open problem                                                 |
| G4     | Perpetual discovery rate bound                   | 06_fixed_point, Proposition SE-1.2 | <b>CONJECTURED</b>                        | <b>RED</b> needed for main convergence result                                                                   |

| Gap ID | Description                                                   | Location                    | Severity                                         | Resolution                                                                         |
|--------|---------------------------------------------------------------|-----------------------------|--------------------------------------------------|------------------------------------------------------------------------------------|
| G5     | Thm 3’s $K > 2$ construction uses “completely random experts” | 03_unidentified, Appendix A | <b>MINOR</b> (construction is valid but extreme) | Acknowledged as extreme case; $K = 2$ construction is the practically relevant one |

## 6. Final Verdict

### 6.1 Audit Results

| Criterion                    | Result                                                                                                                     |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Circular dependencies</b> | <b>NONE FOUND.</b> The dependency graph is a strict DAG.                                                                   |
| <b>Hidden assumptions</b>    | <b>NONE FOUND.</b> All assumptions are explicitly stated and justified.                                                    |
| <b>Notation conflicts</b>    | <b>3 MODERATE conflicts</b> identified ( $\mathcal{S}/S_t$ , $\delta$ , $\Phi$ ). All documented. None are proof-breaking. |

| Criterion                               | Result                                                                                                           |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Broken proof chains</b>              | <b>NONE.</b> All strong dependencies are verified intact.                                                        |
| <b>Unproven claims marked as proven</b> | <b>1 RETRACTED</b> (SE-1.5, $1/\sqrt{N_t}$ rate — DEFECT-05). Corrected version now acknowledges non-resolution. |
| <b>Gaps acknowledged</b>                | <b>5 gaps</b> documented in Section 5. All are either resolved, minor, or explicitly marked as conjectured.      |

## 6.2 Readiness for arXiv

The proof chain is **structurally sound** for arXiv submission with the following caveats:

1. **Theorem 12.5 (reference-set replay)** should be the primary Lyapunov result. The impossibility result (Theorem 12.2) provides strong motivation for the reference-set replay mechanism.
2. **Notation conflicts** should be resolved before submission (see Section 3.2 recommendations).
3. **Gap G2** (infinite  $\mathcal{X}$  functional convergence) should be explicitly acknowledged as an open problem.
4. **Thm 3 Appendix A** ( $K > 2$  construction) should include the caveat that the construction uses extreme “completely random experts” and

that the  $K = 2$  case is the practically relevant one. This is already documented.

### 6.3 Theorem Status Summary

| Theorem                                            | Status        | Proof Completeness | Key Conditions                        |
|----------------------------------------------------|---------------|--------------------|---------------------------------------|
| <b>Theorem 1</b> (Noise Detection)                 | <b>PROVEN</b> | 100%               | A1-A6                                 |
| <b>Theorem 2</b> (Weak Feature)                    | <b>PROVEN</b> | 100%               | $\delta$ -weak $\phi$ , state balance |
| <b>Theorem 3</b> (Unidentifiability)               | <b>PROVEN</b> | 100%               | Constructive counterexample           |
| <b>Proposition 3</b> (State-Conditioned Weighting) | <b>PROVEN</b> | 100%               | Gibbs inequality                      |
| <b>Proposition 4</b> (Compression Fidelity)        | <b>PROVEN</b> | 100%               | A1-A4<br>(compression-specific)       |

| Theorem                                                        | Status                           | Proof Completeness | Key Conditions                |
|----------------------------------------------------------------|----------------------------------|--------------------|-------------------------------|
| <b>Theorem 12.5</b><br>(Lyapunov Descent with Replay)          | <b>PROVEN</b>                    | 100%               | C1'-C9 + reference-set replay |
| <b>Theorem 12.2</b><br>(Lyapunov Impossibility without Replay) | <b>PROVEN</b>                    | 100%               | Proves necessity of replay    |
| <b>Spring-1 (SE-1)</b><br>(Convergence)                        | <b>PROVEN</b><br>(with Thm 12.5) | 95%                | C1-C9 + Thm 12.5              |
| <b>Spring-2 (SE-2)</b><br>(Completeness)                       | <b>PROVEN</b>                    | 100%               | Physical constraints + SE-1   |
| <b>Regime 3</b><br>(Perpetual Discovery)                       | <b>CONJECTURED</b>               | 0%                 | Open problem                  |
| <b>Phase Diagram</b>                                           | <b>CONJECTURED</b>               | 0%                 | Open problem                  |

*End of PROOF\_CHAIN\_AUDIT.md*

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## References (Audit-Specific)

1. All theorem documents in /theory/theorems/ and /theory/theorems/polished/
2. All self-evolution documents in /theory/self\_evolution/
3. Notation document: /theory/theorems/polished/00\_notation\_and\_dependencies.md
4. Definitions: /theory/definitions/01\_state\_conditioned\_risk.md
5. Propositions: /theory/propositions/03\_state\_conditioned\_weighting.md,  
/theory/propositions/04\_compression\_fidelity.md